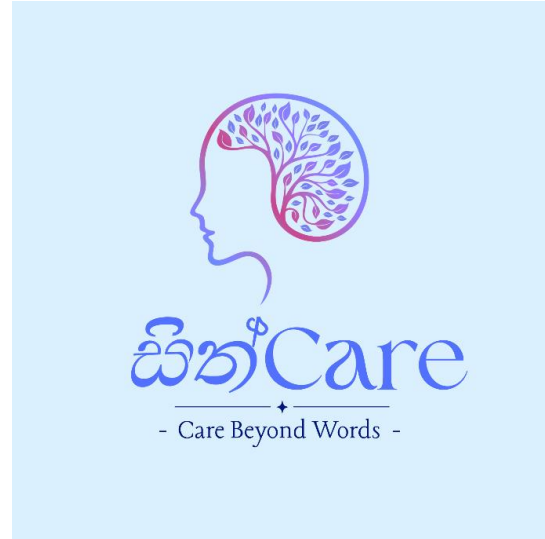




SENG 31242 – System Design Project

SithCare – Core Class Diagram

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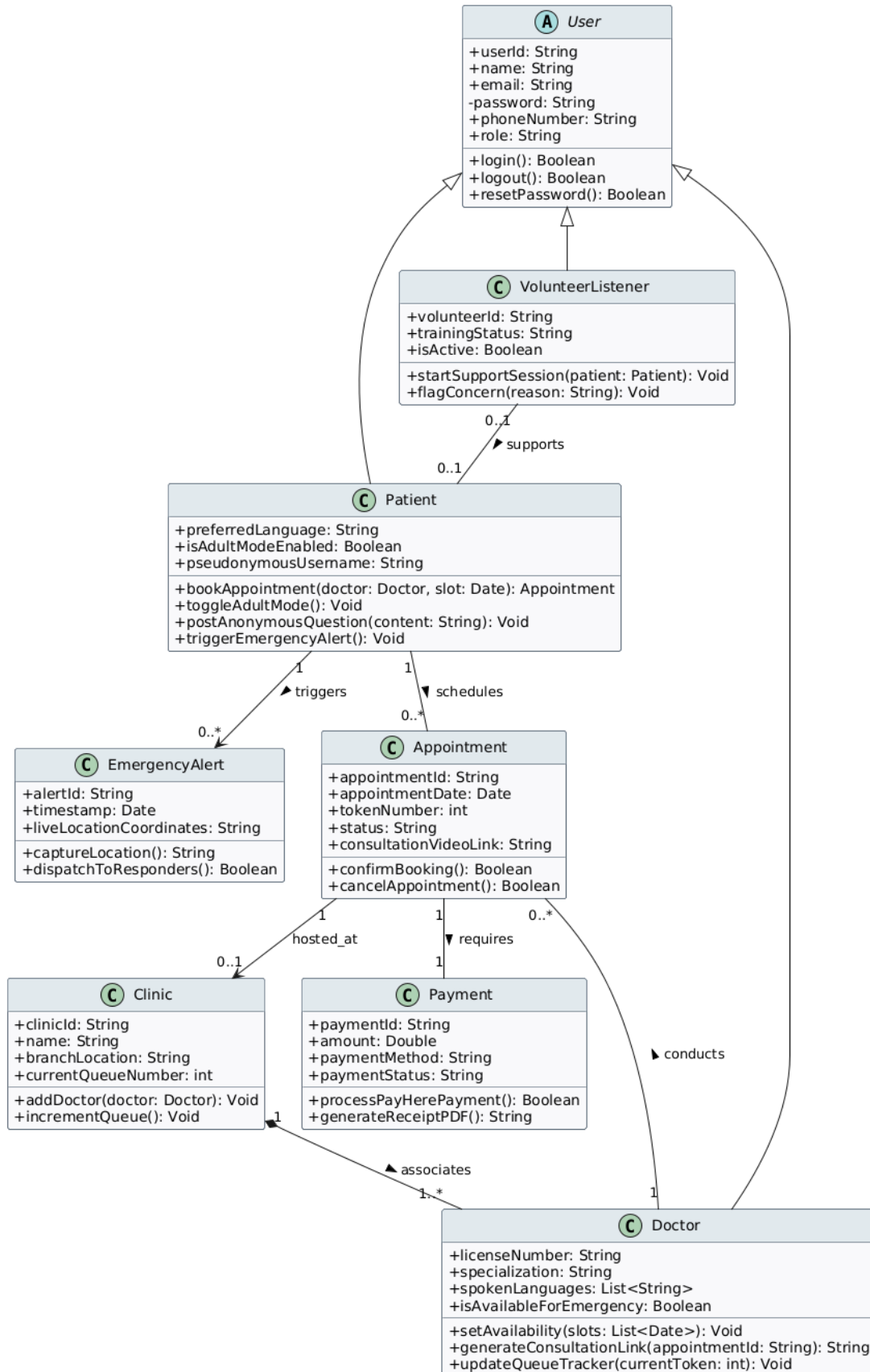
Detailed Design

Core Class Specification

The structural architecture of the **SithCare** platform is realized using an Object-Oriented design paradigm to guarantee strict encapsulation, systemic maintainability, and extensibility. The domain model translates the functional boundaries established in the requirements engineering phase into concrete software entities.

Core Class Diagram

The complete visual layout of the structural system domain is detailed below:



Structural Dictionary & Data Elements

The attributes, method signatures, and visibilities of the components within the design domain are defined below:

1. User (Abstract Base Class)

- **Purpose:** Serves as the generalized superclass for all system actors to maintain identity management and implement localized Role-Based Access Control (RBAC).
- **Attributes:**
 - + `userId`: String — Unique system token identifying the account.
 - + `name`: String — Legal name profile string.
 - + `email`: String — Unique electronic mail address.
 - - `password`: String — Encrypted cryptographic credential signature.
 - + `phoneNumber`: String — Primary telephone contact context.
 - + `role`: String — System authorization flag (e.g., Patient, Doctor, Volunteer).
- **Methods:**
 - + `login()`: Boolean — Authenticates user credentials.
 - + `logout()`: Boolean — Terminates active state session context.
 - + `resetPassword()`: Boolean — Initiates the secure recovery sequence.

2. Patient

- **Purpose:** Models the customer interaction domain, facilitating access to localized channeling features and privacy protection states.
- **Attributes:**
 - + `preferredLanguage`: String — Local language preference selection (Sinhala, Tamil, English).
 - + `isAdultModeEnabled`: Boolean — State flag toggling simplified menu layout mechanics.
 - + `pseudonymousUsername`: String — Pseudonym mask to guarantee community interaction confidentiality.
- **Methods:**

- + bookAppointment(doctor: Doctor, slot: Date): Appointment — Schedules an interface slot with a doctor entity.
- + toggleAdultMode(): Void — Modifies interface layout complexity constraints.
- + postAnonymousQuestion(content: String): Void — Publishes open-forum forum text data safely.
- + triggerEmergencyAlert(): Void — Instantly dispatches system telemetry to immediate responders.

3. Doctor

- **Purpose:** Models specialized professional system personas managing scheduling parameters and session consultation domains.
- **Attributes:**
 - + licenseNumber: String — Professional registry identifier validation token.
 - + specialization: String — Area of clinical focus domain (e.g., Psychiatry, Psychology).
 - + spokenLanguages: List<String> — Collection of supported local communication tracks.
 - + isAvailableForEmergency: Boolean — Priority tracking flag indicating same-day emergency triage availability.
- **Methods:**
 - + setAvailability(slots: List<Date>): Void — Configures the visible scheduling allocation board.
 - + generateConsultationLink(appointmentId: String): String — Configures a secure workspace link for telemedicine execution.
 - + updateQueueTracker(currentToken: int): Void — Increments visual progress numbers dynamically to optimize on-site flow.

4. VolunteerListener

- **Purpose:** Captures the data structures governing certified non-clinical emotional support interactions.
- **Attributes:**
 - + volunteerId: String — Unique system tracking index.
 - + trainingStatus: String — Certification verification indicator.

- + isActive: Boolean — Current active duty tracking flag.
- **Methods:**
 - + startSupportSession(patient: Patient): Void — Initiates an entry point session trace for non-judgmental dialogue.
 - + flagConcern(reason: String): Void — Escalates high-risk indicators directly to administrative or clinic levels.

5. Appointment

- **Purpose:** Acts as the coordinating transaction block connecting actors, scheduling constraints, and clinical environments.
- **Attributes:**
 - + appointmentId: String — Core transaction identifier.
 - + appointmentDate: Date — Specific scheduled point structure.
 - + tokenNumber: int — Position number tracker for physical arrival management.
 - + status: String — Operational status tracking parameter (e.g., Confirmed, Completed, Cancelled).
 - + consultationVideoLink: String — Active secure telemedicine uniform resource locator.
- **Methods:**
 - + confirmBooking(): Boolean — Registers the structural booking instance firmly.
 - + cancelAppointment(): Boolean — Processes localized drop sequences and fires out notifications.

6. Payment

- **Purpose:** Manages financial operations and maintains compliance with the localized clearing networks.
- **Attributes:**
 - + paymentId: String — Audit record validation code.
 - + amount: Double — Denominated pricing scale tracked strictly in Sri Lankan Rupees (LKR).
 - + paymentMethod: String — Execution flag indicating PayHere digital or manual on-site methods.

- + paymentStatus: String — Tracking context (e.g., Pending, Settled, Refunded).
- **Methods:**
 - + processPayHerePayment(): Boolean — Contacts the PayHere third-party gateway payment API.
 - + generateReceiptPDF(): String — Formats and structures an official transactional receipt artifact.

7. Clinic

- **Purpose:** Manages physical facility profile spaces and local queue allocations within multi-branch frameworks.
- **Attributes:**
 - + clinicId: String — Unique enterprise facility identifier.
 - + name: String — Public title designation.
 - + branchLocation: String — Geographic district string mapping the local context.
 - + currentQueueNumber: int — Monitored marker capturing live progress indicators.
- **Methods:**
 - + addDoctor(doctor: Doctor): Void — Formally associates a specialized practitioner to the facility profile.
 - + incrementQueue(): Void — Modifies the line metrics value dynamically to prevent cluster confusion.

8. EmergencyAlert

- **Purpose:** Captures and dispatches critical emergency data when severe psychological crises manifest.
- **Attributes:**
 - + alertId: String — Critical alert track tag.
 - + timestamp: Date — Logged generation marker.
 - + liveLocationCoordinates: String — Active GPS telemetry fetched strictly through explicit user permissions.
- **Methods:**

- + captureLocation(): String — Validates and logs current coordinate telemetry data strings securely.
- + dispatchToResponders(): Boolean — Routes priority information to regional helplines or responders immediately.

4.3 Behavioral Multiplicities & Relationship Rationale

The dynamic behavioral couplings indicated by structural notations are evaluated below:

- **Generalization (Inheritance):** User acts as an abstract foundation class. Generalized properties are cleanly passed to Patient, Doctor, and VolunteerListener child objects, enabling standardized verification workflows while optimizing code reuse.
- **Composition (\$1 \rightarrow 1..*\$):** A strict composition dependency is modeled from Clinic to Doctor. A doctor profile cannot exist standalone within the institutional context without a concrete enterprise facility association, mapping cleanly to multi-clinic configuration requirements.
- **Directed Association (\$1 \rightarrow 0..*\$):** Patient coordinates zero-to-many schedules with Appointment entities. Similarly, a specialized Doctor conducts zero-to-many scheduled appointments.
- **Structural Association (\$1 \rightarrow 1\$):** Each initialized Appointment forces a precise one-to-one compliance binding with a Payment transaction block, guaranteeing that no scheduling event passes without creating an audited monetary trace.
- **Conditional Association (\$0..1 \rightarrow 0..1\$):** A highly isolated supports tracking link matches a VolunteerListener to a Patient profile, ensuring that conversational peer interactions are transiently mapped without breaching database privacy constraints.